

RBE 500 – Homework 5

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Part 1-A: Velocity Kinematics of ABB IRB1400

Same DH parameters as HW3 and HW4:

Joint i	θ_i	d_i (mm)	a_i (mm)	α_i
1	q_1	475	150	$+90^\circ$
2	$q_2 + 90^\circ$	0	600	0°
3	q_3	0	120	$+90^\circ$
4	$q_4 + 90^\circ$	720	0	$+90^\circ$
5	q_5	0	0	-90°
6	q_6	85	0	0°

Question 1: 6×6 Manipulator Jacobian

Since all six joints are revolute, each column of J is:

$$J_i = \begin{bmatrix} z_{i-1}^0 \times (p_e^0 - p_{i-1}^0) \\ z_{i-1}^0 \end{bmatrix}$$

where z_{i-1}^0 is the z -axis of frame $i - 1$ in the base frame, p_e^0 is the EE position, and p_{i-1}^0 is the origin of frame $i - 1$.

Column 1 is the simplest since $z_0^0 = [0, 0, 1]^T$:

$$J_{l1} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \begin{bmatrix} p_{ex} \\ p_{ey} \\ p_{ez} \end{bmatrix} = \begin{bmatrix} -p_{ey} \\ p_{ex} \\ 0 \end{bmatrix}, \quad J_{a1} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

The remaining columns follow the same pattern. Applying the trig identities $\sin(q + 90^\circ) = \cos q$ and $\cos(q + 90^\circ) = -\sin q$ to simplify the DH joint offsets, the z -axes work out to:

Axis	Expression
z_0	$[0, 0, 1]^T$
z_1	$[\sin q_1, -\cos q_1, 0]^T$
z_2	$[\sin q_1, -\cos q_1, 0]^T \quad (\alpha_2 = 0, \text{ so } z_2 \parallel z_1)$
z_3	$[c_1 c_{23}, s_1 c_{23}, -s_{23}]^T \quad (c_{23} = \cos(q_2 + q_3), s_{23} = \sin(q_2 + q_3))$
z_4	$[-c_1 c_{23} s_4 - s_1 c_4, -s_1 c_{23} s_4 + c_1 c_4, s_{23} s_4]^T$
z_5	$[c_1 c_{23} c_4 s_5 - s_1 s_4 s_5 + c_1 s_{23} c_5, \dots]^T$

The lower half of J is just $J_a = [z_0 \mid z_1 \mid z_2 \mid z_3 \mid z_4 \mid z_5]$, and the upper half uses the cross-product formula with these axes and the EE position from T_6^0 .

The following MATLAB code builds and evaluates J numerically:

Listing 1: MATLAB: 6×6 Jacobian for ABB IRB1400

```

1 syms q1 q2 q3 q4 q5 q6 real
2
3 dh = @(th,d,a,al) ...
4     [cos(th) -sin(th)*cos(al) sin(th)*sin(al) a*cos(th);
5      sin(th) cos(th)*cos(al) -cos(th)*sin(al) a*sin(th);
6      0 sin(al) cos(al) d;
7      0 0 0 1];
8
9 d1=475; a1=150; a2=600; a3=120; d4=720; d6=85;
10
11 T01 = dh(q1, d1,a1, pi/2); T12 = dh(q2+pi/2, 0,a2, 0);
12 T23 = dh(q3, 0,a3, pi/2); T34 = dh(q4+pi/2,d4, 0, pi/2);
13 T45 = dh(q5, 0, 0,-pi/2); T56 = dh(q6, d6, 0, 0);
14
15 T02=T01*T12; T03=T02*T23; T04=T03*T34; T05=T04*T45; T06=T05*T56;
16
17 z0=[0;0;1]; p0=[0;0;0];
18 z1=T01(1:3,3); p1=T01(1:3,4);
19 z2=T02(1:3,3); p2=T02(1:3,4);
20 z3=T03(1:3,3); p3=T03(1:3,4);
21 z4=T04(1:3,3); p4=T04(1:3,4);
22 z5=T05(1:3,3); p5=T05(1:3,4);
23 pe=T06(1:3,4);
24
25 J = sym(zeros(6,6));
26 J(:,1)=[cross(z0,pe-p0);z0]; J(:,2)=[cross(z1,pe-p1);z1];
27 J(:,3)=[cross(z2,pe-p2);z2]; J(:,4)=[cross(z3,pe-p3);z3];
28 J(:,5)=[cross(z4,pe-p4);z4]; J(:,6)=[cross(z5,pe-p5);z5];
29
30 % Evaluate numerically at HW4 Part 1.3 configuration

```

```

31 q_vals = [11.31, 24.96, -1.57, 0, 0, 0] * pi/180;
32 J_num = double(subs(J, [q1,q2,q3,q4,q5,q6], q_vals));
33 disp(round(J_num, 4));

```

The symbolic Jacobian entries are too large to print (each element is several lines of nested trig), so the numerical result at the HW4 configuration is shown below in Figure 1.

Full numerical Jacobian J:

-115.3206	-954.7672	-421.3691	0	-16.6700	0
576.5994	-190.9546	-84.2743	0	83.3493	0
0	438.0185	691.2097	0	0	0
0	0.1961	0.1961	0.9000	-0.3893	0.9000
0	-0.9806	-0.9806	0.1800	-0.0779	0.1800
1.0000	0	0	0.3970	0.9178	0.3970

Figure 1: MATLAB output: 6×6 Jacobian evaluated numerically at $\mathbf{q} = [11.31^\circ, 24.96^\circ, -1.57^\circ, 0^\circ, 0^\circ, 0^\circ]$.

Question 2: Joint Velocities

Fixing the wrist at zero ($q_4 = q_5 = q_6 = 0$) and using the joint angles from HW4 Part 1.3:

$$\mathbf{q} = [11.31^\circ, 24.96^\circ, -1.57^\circ, 0^\circ, 0^\circ, 0^\circ]$$

The full numerical Jacobian at this configuration is shown in Figure 1. With the wrist fixed, only joints 1–3 affect the EE linear velocity, so we pull out the top-left 3×3 block and solve:

$$\dot{\mathbf{x}} = J_l^{(3)} \dot{\mathbf{q}}_{1:3} \Rightarrow \dot{\mathbf{q}}_{1:3} = \left(J_l^{(3)}\right)^{-1} \dot{\mathbf{x}}$$

$$J_l^{(3)} = \begin{bmatrix} -115.32 & -954.77 & -421.37 \\ 576.60 & -190.95 & -84.27 \\ 0 & 438.02 & 691.21 \end{bmatrix}, \quad \det(J_l^{(3)}) = 2.851 \times 10^8 \neq 0$$

$$\left(J_l^{(3)}\right)^{-1} \begin{bmatrix} 5 \\ 5 \\ 10 \end{bmatrix} = \begin{bmatrix} 0.0067 \\ -0.0173 \\ 0.0254 \end{bmatrix} \text{ rad/s} \Leftrightarrow \begin{bmatrix} 0.38 \\ -0.99 \\ 1.46 \end{bmatrix} \text{ deg/s}$$

Verification: $J_l^{(3)} \dot{\mathbf{q}}_{1:3} = [5.00, 5.00, 10.00]^T \text{ mm/s} \checkmark$

Question 3: Singular Configurations

Remove the wrist and set $d_6 = 0$, $q_4 = q_5 = q_6 = 0$, leaving a 3-DOF arm. Per the hint, use only the linear half of J , which becomes the 3×3 matrix J_l . The arm is singular when $\det(J_l) = 0$.

Shoulder Singularity

Column 1 of J_l is:

$$z_0 \times (p_e - p_0) = [0, 0, 1]^T \times [p_{ex}, p_{ey}, p_{ez}]^T = [-p_{ey}, p_{ex}, 0]^T$$

This goes to zero when $p_{ex} = p_{ey} = 0$, i.e. the EE is directly above the base on the z_0 axis. At that point joint 1 spinning has no effect on EE position, so $\det(J_l) = 0$.

The horizontal reach is $R(q_2, q_3) = a_1 + a_2 \sin(q_2 + 90^\circ) + a_3 \cos(q_2 + 90^\circ + q_3) + d_4 \sin(q_2 + 90^\circ + q_3)$, and the singularity occurs whenever $R = 0$. Solving numerically gives one example: $q_1 = 0^\circ$, $q_2 \approx -143.47^\circ$, $q_3 = 0^\circ$, at which $\det(J_l) = 0$ and $\text{rank}(J_l) = 2$.

Elbow Singularity

Because this robot has both $a_3 = 120$ mm and $d_4 = 720$ mm, the effective elbow link is $L = \sqrt{a_3^2 + d_4^2} = 729.9$ mm at internal angle $\phi = \arctan(d_4/a_3) = 80.5^\circ$. The elbow singularity does not fall at simple angles like $q_3 = 0^\circ$ or 180° — it happens wherever the arm hits its workspace boundary.

The symbolic determinant (computed in MATLAB) depends only on q_2 and q_3 (independent of q_1 by symmetry). After simplification:

$$\det(J_l) \propto 600 \sin(q_2 + q_3) - 120 \cos(q_2 + q_3) - 600 \sin q_2 \sin(q_2 + q_3) + 120 \sin q_2 \cos(q_2 + q_3)$$

which can be written as $L \sin(q_2 + q_3 - q_2) \cdot (\dots)$, showing the zeros occur at workspace boundary configurations. Setting this to zero and scanning $q_3 \in [-180^\circ, 180^\circ]$ at $q_1 = 30^\circ$, $q_2 = 45^\circ$ finds four zeros:

$$q_3 \in \{-122.39^\circ, -99.46^\circ, 13.47^\circ, 80.54^\circ\}$$

At each, rank drops to 2. Note that $q_3 = 80.54^\circ = \phi$, which is the arm-straight condition in this DH convention.

Summary:

$$\begin{cases} \textbf{Shoulder:} & R(q_2, q_3) = 0 \quad (\text{EE on } z_0 \text{ axis, e.g. } q_2 \approx -143.5^\circ) \\ \textbf{Elbow:} & q_3 \in \{-122.4^\circ, -99.5^\circ, 13.5^\circ, 80.5^\circ\} \end{cases}$$

For the full 6-DOF robot there is also a wrist singularity at $q_5 = 0^\circ$, where joints 4 and 6 become co-axial.

Part 1-B: Textbook Problems (Spong, 2nd Edition)

Problem 4-15: Cylindrical Manipulator (Figure 3.7)

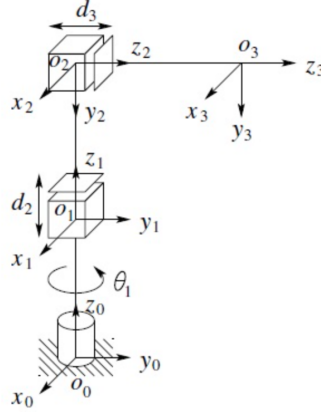


Figure 3.7 Three-link cylindrical manipulator.

Figure 2: Three-link cylindrical manipulator (Spong Figure 3.7). Joint sequence: RPP.

Joint sequence is RPP: joint 1 revolute (θ_1), joints 2 and 3 prismatic (d_2 , d_3).

DH parameters:

Joint i	θ_i	d_i	a_i	α_i
1	q_1	0	0	-90°
2	0	d_2	0	0°
3	0	d_3	0	0°

Axes and origins:

$$\begin{aligned}
 z_0 &= [0, 0, 1]^T, & p_0 &= [0, 0, 0]^T \\
 z_1 &= [0, 0, 1]^T, & p_1 &= [0, 0, 0]^T \\
 z_2 &= [-s_1, c_1, 0]^T, & p_2 &= [0, 0, d_2]^T
 \end{aligned}$$

EE position: $p_e = [-s_1 d_3, c_1 d_3, d_2]^T$.

Jacobian columns:

Joint 1 (revolute):

$$J_{l1} = z_0 \times (p_e - p_0) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \begin{bmatrix} -s_1 d_3 \\ c_1 d_3 \\ d_2 \end{bmatrix} = \begin{bmatrix} -c_1 d_3 \\ -s_1 d_3 \\ 0 \end{bmatrix}, \quad J_{a1} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Joints 2 and 3 (prismatic, so $J_{li} = z_{i-1}$ and $J_{ai} = \mathbf{0}$):

$$J_{l2} = z_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad J_{l3} = z_2 = \begin{bmatrix} -s_1 \\ c_1 \\ 0 \end{bmatrix}$$

Full 6×3 Jacobian:

$$J = \begin{bmatrix} -c_1 d_3 & 0 & -s_1 \\ -s_1 d_3 & 0 & c_1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Singularities: Expanding $\det(J_l)$ along the third row:

$$\det(J_l) = 1 \cdot \det \begin{bmatrix} -c_1 d_3 & -s_1 \\ -s_1 d_3 & c_1 \end{bmatrix} = -d_3(c_1^2 + s_1^2) = -d_3$$

$$\boxed{\det(J_l) = 0 \iff d_3 = 0}$$

The only singularity is when $d_3 = 0$, meaning the EE is sitting on the z_0 rotation axis and joint 1 spinning does nothing.

Problem 4-16: Cartesian Manipulator (Figure 3.17)

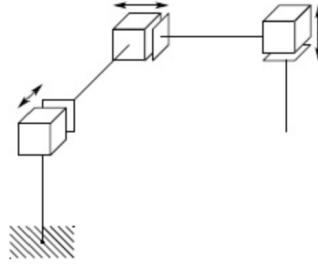


Figure 3.17 Three-link Cartesian robot.

Figure 3: Three-link Cartesian manipulator (Spong Figure 3.17). Joint sequence: PPP.

All three joints are prismatic, translating along fixed orthogonal axes: $z_0 = [0, 0, 1]^T$, $z_1 = [0, 1, 0]^T$, $z_2 = [1, 0, 0]^T$.

For prismatic joints $J_{li} = z_{i-1}$ and $J_{ai} = \mathbf{0}$, so:

$$J = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\det(J_l) = -1 \neq 0$ for all joint values, so this robot has **no singularities**. The three axes are always orthogonal and independent no matter where the joints move.

Problem 4-17: Stanford Manipulator (Completing Example 4.7)

The example gives us o_6 , o_3 , and all the z -axes. The spherical wrist means $o_3 = o_4 = o_5 \triangleq o$, so the vector from the wrist center to the EE is simply $o_6 - o = d_6 z_5$.

The column formulas from the example are:

$$J_i = \begin{bmatrix} z_{i-1} \times (o_6 - o_{i-1}) \\ z_{i-1} \end{bmatrix} \quad (i = 1, 2), \quad J_3 = \begin{bmatrix} z_2 \\ \mathbf{0} \end{bmatrix}, \quad J_i = \begin{bmatrix} z_{i-1} \times (o_6 - o) \\ z_{i-1} \end{bmatrix} \quad (i = 4, 5, 6)$$

Filling these in:

Column 3 (prismatic, $z_2 = [c_1 s_2, s_1 s_2, c_2]^T$):

$$J_3 = \begin{bmatrix} c_1 s_2 \\ s_1 s_2 \\ c_2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Columns 4-6 (wrist, all share origin o , so $o_6 - o = d_6 z_5$):

$$J_{l4} = z_3 \times d_6 z_5, \quad J_{l5} = z_4 \times d_6 z_5, \quad J_{l6} = z_5 \times d_6 z_5 = \mathbf{0}$$

$$J_{a4} = z_3, \quad J_{a5} = z_4, \quad J_{a6} = z_5$$

$J_{l6} = \mathbf{0}$ because joint 6's axis passes through the EE, so spinning it produces no linear velocity.

Complete 6×6 Jacobian:

$$J = \begin{bmatrix} z_0 \times (o_6 - o_0) & z_1 \times (o_6 - o_1) & z_2 & z_3 \times d_6 z_5 & z_4 \times d_6 z_5 & \mathbf{0} \\ z_0 & z_1 & \mathbf{0} & z_3 & z_4 & z_5 \end{bmatrix}$$

Plugging in the expressions from Eq. (4.64)–(4.69) finishes the derivation.

Example 4.7. (Stanford Manipulator).

Consider the Stanford Manipulator of Example 3.3.5 with its associated Denavit–Hartenberg coordinate frames. Note that joint 3 is prismatic and that $o_3 = o_4 = o_5$ as a consequence of the spherical wrist and the frame assignment. Denoting this common origin by o we see that the columns of the Jacobian have the form

$$J_i = \begin{bmatrix} z_{i-1} \times (o_6 - o_{i-1}) \\ z_{i-1} \end{bmatrix} \quad i = 1, 2$$

$$J_3 = \begin{bmatrix} z_2 \\ 0 \end{bmatrix}$$

$$J_i = \begin{bmatrix} z_{i-1} \times (o_6 - o) \\ z_{i-1} \end{bmatrix} \quad i = 4, 5, 6$$

Now, using the A matrices given by Equations (3.18)–(3.20) and the T matrices formed as products of the A matrices, these quantities are easily computed as follows. First, o_j is given by the first three entries of the last column of $T_j^0 = A_1 \cdots A_j$, with $o_0 = (0, 0, 0) = o_1$. The vector z_j is given as $z_j = R_j^0 k$ where R_j^0 is the rotational part of T_j^0 . Thus, it is only necessary to compute the matrices T_j^0 to calculate the Jacobian. Carrying out these calculations one obtains the following expressions for the Stanford Manipulator:

$$o_6 = \begin{bmatrix} c_1 s_2 d_3 - s_1 d_2 + d_6(c_1 c_2 c_4 s_5 + c_1 c_5 s_2 - s_1 s_4 s_5) \\ s_1 s_2 d_3 - c_1 d_2 + d_6(c_1 s_4 s_5 + c_2 c_4 s_1 s_5 + c_5 s_1 s_2) \\ c_2 d_3 + d_6(c_2 c_5 - c_4 s_2 s_5) \end{bmatrix} \quad (4.64)$$

$$o_3 = \begin{bmatrix} c_1 s_2 d_3 - s_1 d_2 \\ s_1 s_2 d_3 + c_1 d_2 \\ c_2 d_3 \end{bmatrix} \quad (4.65)$$

The various z -axes are given as

$$z_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad z_1 = \begin{bmatrix} -s_1 \\ c_1 \\ 0 \end{bmatrix} \quad (4.66)$$

$$z_2 = \begin{bmatrix} c_1 s_2 \\ s_1 s_2 \\ c_2 \end{bmatrix}, \quad z_3 = \begin{bmatrix} c_1 s_2 \\ s_1 s_2 \\ c_2 \end{bmatrix} \quad (4.67)$$

$$z_4 = \begin{bmatrix} -c_1 c_2 s_4 - s_1 c_4 \\ -s_1 c_2 s_4 + c_1 c_4 \\ s_2 s_4 \end{bmatrix} \quad (4.68)$$

$$z_5 = \begin{bmatrix} c_1 c_2 c_4 s_5 - s_1 s_4 s_5 + c_1 s_2 c_5 \\ s_1 c_2 c_4 s_5 + c_1 s_4 s_5 + s_1 s_2 c_5 \\ -s_2 c_4 s_5 + c_2 c_5 \end{bmatrix} \quad (4.69)$$

The Jacobian of the Stanford Manipulator is now given by combining these expressions according to the given formulas (Problem 4–17).

Figure 4: Example 4.7: column templates, EE position o_6 , wrist center o_3 (top), and z -axis expressions z_0 – z_5 from Eq. 4.66–4.69 (bottom).